

ParcelPilot Support Agent — 10-Minute Video Demo Script

Production Evaluation Walkthrough • Exact Speech Prompts • On-Screen Actions

Primary Production URL (Render): <https://parcelpilot-agent-05e4.onrender.com>

Secondary Production URL (Railway): <https://parcelpilot-agent-production.up.railway.app>

GitHub Repository: <https://github.com/Sumitboii/-parcelpilot-agent>

Timeline & Agenda Overview

Timecode	Segment	Core Topic Demonstrated
00:00 – 01:15	1. Introduction & Context	Overview of ParcelPilot, support bottlenecks, and core objectives
01:15 – 02:30	2. Architecture & Multi-Key Pool	Plain Python tool router, multi-key health step function, zero-latency LRU
02:30 – 04:15	3. Demo 1: Source Hierarchy	Northstar cancellation query (Agreement \$2 overrides Policy & bad TKT-450)
04:15 – 05:45	4. Demo 2: Confirmation Gate	TKT-505 P1 incident (SLA breach check + interactive confirmation card)
05:45 – 07:00	5. Demo 3: Service Credit	LumenWorks delay credit (Custom agreement ₹300 fixed override)
07:00 – 08:15	6. Demo 4: Known Issues	KI-208 bulk upload limit vs bug workaround & Policy v2 refusal
08:15 – 09:15	7. Demo 5: Live Issues Radar	Live issues sidebar (SLA breaches, KI clusters, account surges with 1-click trigger)
09:15 – 10:00	8. Trust, Metrics & Wrap-Up	Grounded citations, 66/66 test suite, hallucination prevention

Detailed Script: Words to Speak & Screen Actions

Segment 1: Introduction & Context

00:00 – 01:15

ON-SCREEN ACTIONS:

- Share screen showing the clean **ParcelPilot Support Agent** UI (<https://parcelpilot-agent-05e4.onrender.com>).
- Highlight the top bar with active user role (Rohit – Support Agent), the central conversation area, and the left **Live Issues** sidebar.

WORDS TO SPEAK:

“Hello everyone! Today I’m excited to present the **ParcelPilot AI Support Agent** — an intelligent operational copilot built for ParcelPilot’s support and customer operations teams.

ParcelPilot coordinates logistics across multiple carriers, processing hundreds of complex enterprise queries every day. Today, support agents lose hours manually cross-referencing PDFs, custom enterprise contracts, order databases, and ticketing systems.

The core challenge isn’t just fetching data — it’s handling **conflicts, speed, and trust**. Customer agreements override standard policies, old policies get deprecated, rate limits can cause unexpected downtime, and historical ticket resolutions often contain errors.

We built this production system to solve that exact problem: delivering sub-second, trustworthy, cited answers with automated multi-key failover while keeping humans strictly in control of critical business actions.”

Segment 2: Architecture & Multi-Key Health Step Function

01:15 – 02:30

ON-SCREEN ACTIONS:

- Briefly show the top-right role selector, the clean streaming message area, and point out the sub-second streaming responsiveness.
- Open the `/docs` or mention the `/keys/health` endpoint briefly.

WORDS TO SPEAK:

“Let’s take a look at the architecture under the hood.

First, **Multi-Key Health Step Function**: To guarantee zero downtime and eliminate rate-limit disruptions, we built a custom `GroqKeyPoolManager`. It automatically discovers all available Groq API keys, dispatches requests with zero added latency using an in-memory Least-Recently-Used rotation, and includes an automated step function that fails over to a backup key in milliseconds if a 429 rate limit or 413 token limit occurs. A background worker continuously probes key health every 30 seconds.

Second, **Strict Source Authority Hierarchy**: The agent is hard-wired with a deterministic hierarchy:

1. Signed customer agreements always take top priority.
2. Current support policies (Policy v3 CURRENT) come second.
3. Standard SOPs and operations guides come third.
4. Historical ticket resolutions are treated strictly as unverified context.
5. Any deprecated documents — such as Support Policy v2 — are completely filtered out.

Third, **Safety & Privacy**: Every data query passes through an in-memory data store with Role-Based Access Control, and state-changing actions like escalations are intercepted by an interactive confirmation gate.”

Segment 3: Scenario 1 — Source Authority & Contract Override

02:30 – 04:15

ON-SCREEN ACTIONS:

- Click the suggestion chip: **Northstar cancellation** (or type: *'Can Northstar cancel ORD-1001 without a fee? Explain why.'*)
- Press **Send**.
- Point out the tool chips appearing (`data_lookup` , `document_search`) and the rapid token streaming.

WORDS TO SPEAK:

“Let’s test our first scenario: a contract conflict and a known bad historical record.

I’ll ask: *'Can Northstar cancel ORD-1001 without a fee?'*

Notice what happened:

1. The agent immediately called `data_lookup` to check `ORD-1001`. It saw the order is still in `BOOKED` status.
2. It then queried `document_search` for Northstar’s specific enterprise agreement.

Look at the answer:

- The agent confirms: **Yes, Northstar can cancel with zero fee.**
- It cites **Section 2 of Northstar’s Enterprise Agreement**, which explicitly waives all cancellation fees for shipments cancelled before pickup.
- Most importantly, it actively highlights that a past ticket resolution — **TKT-450** — incorrectly told the customer they owed ₹250. The agent correctly rejected that historical mistake in favor of the authoritative contract.
- Every single claim is backed by a clickable citation badge.”

Segment 4: Scenario 2 — Human-in-the-Loop Confirmation Gate

04:15 – 05:45

ON-SCREEN ACTIONS:

- Click the suggestion chip: **TKT-505 API key** (or type: 'What should I do about TKT-505?')
- Press **Send**.
- Show the tool chips, the policy check, and then the interactive ⚡ **Confirm Action** card rendered on screen.
- Click **Confirm** button and show the generated escalation ID (ESC-20260816-XXXX).

WORDS TO SPEAK:

"Now let's look at a critical operational safety requirement: **The Confirmation Gate**.

I'm asking about **TKT-505**, a ticket reporting possible API key exposure for Axis Labs.

Notice how the agent reasons through this:

1. It checks the ticket: Opened at 08:30, snapshot is 11:00 — so 2 hours and 30 minutes have elapsed.
2. It checks policy: Axis Labs is Enterprise with no custom agreement, so Standard Policy v3 applies. P1 target is 30 minutes.
3. The SLA is breached by 2 hours. Because suspected credential leak is classified as **P1**, policy mandates immediate escalation.

But instead of executing the escalation silently in the background, look at the screen: The agent rendered an interactive **Confirmation Card** with pre-filled metadata: Ticket ID, Severity, Reason, and Assignee.

Until I click 'Confirm', nothing is written to our audit database. When I click **Confirm**, it formally logs the escalation and returns a permanent tracking ID."

Segment 5: Scenario 3 — Service Credit Calculations

05:45 – 07:00

ON-SCREEN ACTIONS:

- Click the suggestion chip: **LumenWorks credit** (or type: 'Is ORD-2002 eligible for a service credit?')
- Press **Send**.
- Highlight the calculation breakdown, delay duration, and contract override.

WORDS TO SPEAK:

"Next, let's test mathematical calculation and custom SLA terms with **ORD-2002** for LumenWorks.

The standard ParcelPilot SOP specifies that a credit is eligible after a 2-hour delay using a 10% formula.

But watch what the agent does:

1. It looks up ORD-2002 : the pickup window closed at 06:30, making the delay **4 hours and 30 minutes** due to carrier fault.
2. It retrieves LumenWorks's Service Agreement.
3. It applies LumenWorks's custom contract clause: their threshold is 4 hours, and instead of a variable percentage, it awards a **fixed ₹300 credit**.
4. It also confirms that because ₹300 is below the ₹1,000 policy threshold, manager pre-approval is not required."

Segment 6: Scenario 4 — Known Issues & Guardrails

07:00 – 08:15

ON-SCREEN ACTIONS:

- Click the suggestion chip: **Bulk upload limit** (or type: *'LumenWorks 4200-row CSV fails to upload'*)
- Press **Send**.
- Show how the agent distinguishes product specs from workaround limits.

WORDS TO SPEAK:

"Let's look at how the agent handles product bugs versus real specifications.

When LumenWorks asks why their 4,200-row CSV upload failed:

- The agent clarifies that the official product specification is **5,000 rows**.
- But it detects known bug **KI-208**, which causes intermittent failures above 3,000 rows.
- It provides the exact operational workaround: split the file into smaller batches under 3,000 rows while engineering deploys a fix.
- Again, it flags that old ticket **TKT-451** incorrectly told the customer 3,000 was the plan limit, setting the record straight."

Segment 7: Scenario 5 — Interactive Live Issues Radar

08:15 – 09:15

ON-SCREEN ACTIONS:

- Move the cursor to the left **Live Issues** sidebar.
- Click on **TKT-504** or **Account Cluster (Northstar)** directly from the sidebar.
- Point out that clicking the item immediately triggers the live investigation with full AI streaming and tool chips without manual typing.

WORDS TO SPEAK:

"Now let's look at **Proactive Issue Detection & Investigation**.

Reactive chat only helps after a human asks. Operations teams need to know what's on fire *before* customers escalate.

On the left sidebar, our backend runs a continuous proactive sweep across all active orders and tickets:

-  **SLA Breaches:** Instantly surfaces breached tickets like TKT-505 and TKT-501 with exact elapsed times.
-  **Known Issue Clusters:** Groups tickets linked to active bugs like KI-208 and KI-211.
-  **Account Clusters:** Detects when a single enterprise account has multiple complaints within 7 days.

Watch this: I can simply click **TKT-504** or **Account Cluster** directly on the sidebar. The UI immediately begins the investigation, runs the tools, and explains the correlation between the orders in real time."

ON-SCREEN ACTIONS:

- Scroll up through the chat transcript showing clean Markdown tables, citation badges, and confirmation records.
- Switch user role to **Priya Mehta – CSM** to demonstrate role-based session isolation.
- Show the test suite passing summary (66/66 tests passed).

WORDS TO SPEAK:

“To wrap up:

- **Trust & Grounding:** Every single factual statement cites an authoritative document or database key. The agent never hallucinates.
- **Speed & Resilience:** Powered by a multi-key Groq pool with zero-latency failover and a lightweight, memory-efficient in-memory vector store (< 40MB RAM).
- **Verified Quality:** 66 out of 66 automated tests pass across unit tests, key rotation, and full evaluation scenarios.
- **Primary North Star Metric:** We measure **Resolution Accuracy & Groundedness (% of responses with zero hallucinations and verified source hierarchy resolution)**.

Thank you for your time, and I look forward to your questions!”